

Mathematical Analysis Lecture 25. 定积分.

定理: $f(x)$ 在 $[a, b]$ 可积的充要条件是: 任意划分 $P, \lambda = \max \{\Delta x_i\} \rightarrow 0, \lim_{\lambda \rightarrow 0} \bar{S}(P) = L = I = \lim_{\lambda \rightarrow 0} \underline{S}(P)$

充分性: 由于: $\underline{S}(P) \leq \sum_{i=1}^n f(\xi_i) \Delta x_i \leq \bar{S}(P)$ 由夹逼定理显然.

必要性: 特殊地, 取 ξ_i 是 $[x_{i-1}, x_i]$ 中满足 $0 \leq M_i - f(\xi_i) < \frac{\varepsilon}{2(b-a)}$ M_i 为上确界故 ξ_i 一定取得到

$$\therefore |\bar{S}(P) - \sum_{i=1}^n f(\xi_i) \Delta x_i| = \sum_{i=1}^n (M_i - f(\xi_i)) \Delta x_i = \frac{\varepsilon}{2(b-a)} \cdot (b-a) = \frac{\varepsilon}{2}$$

$$|\bar{S}(P) - I| \leq |\bar{S}(P) - \sum_{i=1}^n f(\xi_i) \Delta x_i| + |\sum_{i=1}^n f(\xi_i) \Delta x_i - I| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

等价表述: 记 $w_i = M_i - m_i$ 为 $f(x)$ 在 $[x_{i-1}, x_i]$ 上的振幅, 对 P 划分, 当 $\lambda = \max \{\Delta x_i\} \rightarrow 0$ 时, $\lim_{\lambda \rightarrow 0} \sum_{i=1}^n w_i \cdot \Delta x_i = 0$

→ 推论 1: 连续函数 (闭区间) 一定可积.

证 → 一致连续: 设 $f(x)$ 在 $[a, b]$ 上连续, 则 $|f(x') - f(x'')| < \frac{\varepsilon}{b-a}, |x' - x''| < \delta$

对任意划分: 只要 $\lambda = \max(\Delta x_i) < \delta$, 则 $w_i = \max f(x) - \min f(x) < \frac{\varepsilon}{b-a}, x \in [x_{i-1}, x_i]$.

$$\therefore \sum_{i=1}^n w_i \cdot \Delta x_i < \frac{\varepsilon}{b-a} \cdot \sum_{i=1}^n \Delta x_i = \varepsilon, \text{ 则可积}$$

→ 推论 2: 闭区间上的单调函数必可积.

$\therefore \forall [x_{i-1}, x_i], w_i = f(x_i) - f(x_{i-1})$ 对 $\forall \varepsilon > 0$, 取 $\delta = \frac{\varepsilon}{f(b) - f(a)} > 0$, 当 $\lambda = \max(\Delta x_i) < \delta$,

$$\sum_{i=1}^n \Delta x_i \cdot w_i < \frac{\varepsilon}{f(b) - f(a)} \sum_{i=1}^n (w_i) = \varepsilon \quad \text{QED}$$

等价表述: 有界函数 $f(x)$ 在 $[a, b]$ 上可积的充要条件是: 对 $\forall \varepsilon > 0, \exists P$, 使 $\sum_{i=1}^n w_i \cdot \Delta x_i < \varepsilon$.

→ 推论 3: 闭区间上只有有限个不连续点的有界函数必可积.

定积分的性质: 1. $\int_a^b f(x) dx = - \int_b^a f(x) dx$ ($\int_a^a f(x) dx = 0$) 规定? 还是性质?

$$2. \int_a^b dx = b - a. \quad 3. \int_a^b k f(x) dx = k \int_a^b f(x) dx$$

$$4. \int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

$$5. \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

$$6. \text{若 } f(x) \leq g(x), \text{ 则 } \int_a^b f(x) \leq \int_a^b g(x)$$

$$\text{推论 } |\int_a^b f(x) dx| \leq \int_a^b |f(x)| dx$$